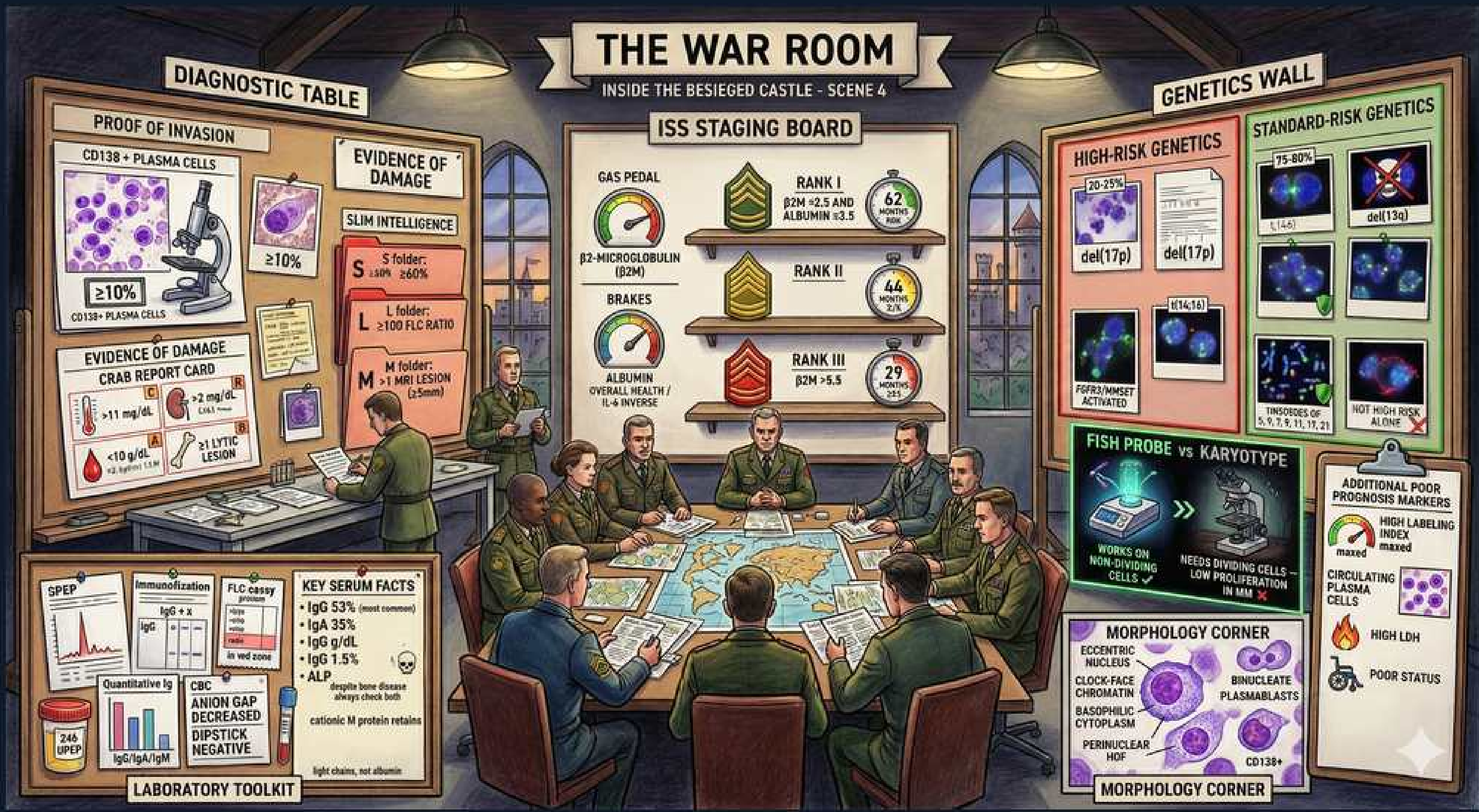
Multiple Myeloma — Diagnosis, Staging (ISS/R-ISS) & Genetics



1. CD138 / ≥10% plasma cells

WHAT YOU SEE

A 'PROOF OF INVASION' card with a CD138 plasma-cell microscope image and '≥10%' on the marrow.

WHAT IT ENCODES

Diagnosis of multiple myeloma requires clonal bone-marrow plasma cells ≥10% (or a biopsy-proven plasmacytoma) PLUS either a CRAB end-organ feature or a myeloma-defining biomarker. CD138 (syndecan-1) is the immunohistochemical marker that identifies plasma cells, and clonality is shown by kappa/lambda light-chain restriction.

BOARD TRAP

Marrow plasma cells ≥10% alone is NOT myeloma — without CRAB or a defining biomarker it is smoldering myeloma; CD138 marks plasma cells but clonality needs light-chain restriction.

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2. M-spike (SPEP / ≥3 g/dL)

WHAT YOU SEE

An 'EVIDENCE OF DAMAGE' card with a sharp monoclonal spike and '≥3 g/dL' serum M-protein.

WHAT IT ENCODES

Serum protein electrophoresis (SPEP) detects a monoclonal spike; immunofixation identifies the heavy/light-chain type (IgG most common, then IgA). MGUS = M-protein <3 g/dL AND marrow plasma cells <10% AND no CRAB. Smoldering myeloma = M-protein ≥3 g/dL or plasma cells 10-60% but no CRAB.

BOARD TRAP

MGUS vs smoldering vs active myeloma is defined by thresholds (M-protein 3 g/dL, plasma cells 10% and 60%) plus presence/absence of CRAB — memorize the cutoffs, they are heavily tested.

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3. Serum free light chains (FLC ratio)

WHAT YOU SEE

A 'SLIM INTELLIGENCE' card showing kappa:lambda free-light-chain ratio with an FLC ratio ≥100 flag.

WHAT IT ENCODES

Serum free light-chain (FLC) assay quantifies kappa and lambda; an abnormal kappa:lambda ratio shows clonality and is essential for light-chain-only (non-secretory) myeloma and AL amyloidosis. An involved:uninvolved FLC ratio ≥100 (with involved FLC ≥100 mg/L) is a myeloma-defining biomarker that upgrades smoldering to active disease.

BOARD TRAP

An FLC ratio ≥100 is a 'SLiM' biomarker that defines active myeloma even WITHOUT CRAB — don't wait for end-organ damage to treat when this is present.

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4. ISS Rank I — β2M <3.5 / Alb ≥3.5

WHAT YOU SEE

An 'ISS STAGING BOARD' rank-I chevron with 'β2-microglobulin <3.5, albumin ≥3.5' and median survival ~62 months.

WHAT IT ENCODES

The International Staging System (ISS) uses just two labs: beta-2-microglobulin and albumin. Stage I = β2M <3.5 mg/L AND albumin ≥3.5 g/dL (best prognosis). Beta-2-microglobulin reflects tumor burden and renal function; low albumin reflects IL-6/inflammatory burden.

BOARD TRAP

ISS uses ONLY beta-2-microglobulin and albumin — it does NOT include calcium, hemoglobin, or creatinine (those are Durie-Salmon). Don't confuse the two staging systems.

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5. ISS Rank II — intermediate

WHAT YOU SEE

A rank-II chevron labeled 'INTERMEDIATE' with median survival ~44 months.

WHAT IT ENCODES

ISS Stage II is the intermediate category: neither stage I nor stage III criteria (e.g., β2M 3.5-5.5 mg/L, or β2M <3.5 with albumin <3.5). Median overall survival historically ~44 months but markedly improved with modern triplet/quadruplet therapy.

BOARD TRAP

Stage II is a 'neither I nor III' category — defined by exclusion, so know stages I and III precisely and II falls out in between.

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6. ISS Rank III — β2M >5.5

WHAT YOU SEE

A rank-III chevron with 'β2M >5.5' and the shortest median survival ~29 months.

WHAT IT ENCODES

ISS Stage III = beta-2-microglobulin >5.5 mg/L (highest tumor burden / worst renal function), the worst prognosis. The Revised ISS (R-ISS) adds high-risk cytogenetics and serum LDH to refine this — R-ISS III requires stage-III β2M PLUS high-risk genetics or high LDH.

BOARD TRAP

R-ISS upgrades ISS by adding LDH and high-risk FISH [del(17p), t(4;14), t(14;16)] — a high β2M alone is ISS III, but R-ISS III also needs adverse genetics or elevated LDH.

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7. High-risk genetics — del(17p), t(4;14)

WHAT YOU SEE

A 'GENETICS WALL' high-risk panel showing del(17p) and t(4;14) FISH probes flagged red.

WHAT IT ENCODES

High-risk cytogenetics by FISH include del(17p) (loss of TP53), t(4;14) (FGFR3/MMSET), t(14;16), t(14;20), gain/amp 1q, and del(1p). These confer shorter survival and influence intensity of therapy. del(17p)/TP53 loss is the single most adverse abnormality.

BOARD TRAP

Karyotype/standard cytogenetics misses most myeloma abnormalities (low proliferative index) — you must use FISH on CD138-selected plasma cells to detect del(17p), t(4;14), etc.

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8. Standard-risk genetics — t(11;14)

WHAT YOU SEE
A 'STANDARD-RISK GENETICS' panel with t(11;14) and hyperdiploidy probes in green.

WHAT IT ENCODES
Standard-risk findings include hyperdiploidy and t(11;14) (CCND1). t(11;14) is notable because it predicts sensitivity to the BCL-2 inhibitor venetoclax — a targeted option in relapsed disease. Standard-risk disease has more favorable outcomes than del(17p)/t(4;14).

BOARD TRAP
t(11;14) is standard-risk AND a positive predictive marker for venetoclax response — uniquely actionable; don't lump it with the high-risk translocations.

9. FISH probe vs karyotype

WHAT YOU SEE
A 'FISH PROBE vs KARYOTYPE' panel noting FISH works on non-dividing cells.

WHAT IT ENCODES
FISH on CD138-purified plasma cells is the standard because myeloma plasma cells have a low proliferative rate and rarely yield informative metaphases for conventional karyotyping. FISH detects the prognostic translocations and deletions regardless of cell division.

BOARD TRAP
Don't order conventional karyotype expecting myeloma cytogenetics — its low yield means FISH (interphase) on selected plasma cells is required.

10. Circulating plasma cells (poor)

WHAT YOU SEE
An 'ADDITIONAL POOR PROGNOSTIC MARKERS' card flagging circulating plasma cells and high LDH.

WHAT IT ENCODES
Circulating clonal plasma cells (plasma cell leukemia when ≥5% of peripheral WBC or ≥0.5 x10^9/L) and elevated LDH mark aggressive, high-proliferation disease with poor prognosis. Plasma cell leukemia is an aggressive variant often with extramedullary disease.

BOARD TRAP
Plasma cells in the PERIPHERAL blood (plasma cell leukemia) is an aggressive, high-risk variant — distinct from the marrow-confined disease of ordinary myeloma.

11. Crab report card (CRAB labs)

WHAT YOU SEE
An 'EVIDENCE OF DAMAGE — CRAB REPORT CARD' listing Ca, renal, anemia, lytic lesion icons.

WHAT IT ENCODES
The CRAB report summarizes end-organ damage that defines symptomatic myeloma: hyperCalcemia (>11), Renal (Cr >2 or CrCl <40), Anemia (Hb <10 or >2 below normal), and Bone lytic lesions. Any one (with clonal plasma cells) mandates treatment.

BOARD TRAP
A single CRAB feature defines active myeloma — you don't need the full tetrad; conversely all CRAB-negative with low burden is MGUS/smoldering, which is observed, not treated.

12. Anion gap decreased / dipstick negative

WHAT YOU SEE
A 'KEY SERUM FACTS' / 'LABORATORY TOOLKIT' panel: anion gap decreased, dipstick negative, immunofixation.

WHAT IT ENCODES
A cationic IgG paraprotein can LOWER the calculated anion gap (extra unmeasured cation). Urine dipstick misses light chains (Bence Jones). Immunofixation is more sensitive/specific than SPEP for confirming and typing the monoclonal protein and for documenting complete response (immunofixation-negative).

BOARD TRAP
A low or negative anion gap can be a clue to a cationic paraprotein (myeloma) — and remember urine dipstick fails to detect light chains, requiring immunofixation/UPEP.

13. Eccentric nucleus / clock-face chromatin

WHAT YOU SEE
A 'MORPHOLOGY CORNER' showing eccentric nucleus, clock-face chromatin, perinuclear hof plasma cells.

WHAT IT ENCODES
Classic plasma-cell morphology: eccentric nucleus, 'clock-face' (cartwheel) chromatin, and a perinuclear pale 'hof' (Golgi). Malignant features include binucleate plasmablasts and high plasma-cell percentage. Plasmablastic morphology portends worse prognosis.

BOARD TRAP
A perinuclear hof and clock-face chromatin identify plasma cells; plasmablastic/immature morphology is an adverse feature, not just a benign reactive change.

14. Slim biomarkers (≥60% / FLC100 / MRI)

WHAT YOU SEE
A 'SLIM INTELLIGENCE' summary linking ≥60% plasma cells, FLC ratio ≥100, and >1 focal MRI lesion.

WHAT IT ENCODES
The 'SLiM-CRAB' myeloma-defining biomarkers (any one upgrades smoldering to active even without CRAB): Sixty percent (≥60%) clonal marrow plasma cells, Light-chain ratio ≥100 (involved FLC ≥100 mg/L), and MRI with >1 focal lesion ≥5 mm. These identify patients who will imminently develop end-organ damage.

BOARD TRAP
'SLiM' criteria let you diagnose and treat active myeloma BEFORE CRAB damage occurs — ≥60% plasma cells, FLC ratio ≥100, or >1 MRI focal lesion each suffice on their own.

15. Whole-body imaging (PET/MRI/LDCT)

WHAT YOU SEE

Imaging cues around the staging board pointing to advanced cross-sectional bone imaging.

WHAT IT ENCODES

Modern guidelines replace plain skeletal survey with whole-body low-dose CT, whole-body MRI, or FDG-PET/CT to detect lytic lesions and focal marrow lesions with far higher sensitivity. MRI defines the >1 focal lesion SLiM biomarker; PET is useful for response/extramedullary disease.

BOARD TRAP

Plain skeletal survey is outdated and insensitive — whole-body low-dose CT, MRI, or PET/CT is now standard for detecting bone disease and the MRI-defined focal-lesion biomarker.